ADITYA PRADEEP KOGRI

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CAREER SUMMARY

Graduate Electrical Engineering student at Arizona State University with hands on experience in **RTL design, logic synthesis, physical design**, and **design verification** across **7nm FinFET and 32nm process technologies**. Completed full **ASIC design flow** from RTL through GDSII signoff, including **synthesis, timing analysis**, and **sign-off flows**. Proficient in **SystemVerilog, Verilog, and C++**, with experience writing **scripted automation** for design analysis and verification workflows. Seeking a **Summer 2026 ASIC Engineer Internship**.

EDUCATION

Master's Degree | Electrical Engineering — **Graduating December 2026**
Arizona State University, Tempe, AZ — GPA: 3.1
Coursework: VLSI Design, ASIC Design Flow, Digital Systems and Circuits, Verilog HDL, Digital Verification and Testing, Python

Bachelor's Degree | Electrical and Electronics Engineering — **August 2024**
University Visveswaraya College of Engineering, Bengaluru, India — GPA: 8.11/10
Coursework: Digital Signal Processing, Analog Integrated Circuits, CMOS Transistors, Microcontrollers

TECHNICAL SKILLS

EDA Tools: Synopsys Design Compiler, **Cadence Innovus**, **Virtuoso**, HSPICE, LTspice, **QuestaSim**, ModelSim
ASIC/VLSI Design: RTL Design, **Logic Synthesis**, **Functional Verification**, **Testbench Development**, **Timing Analysis**, **Physical Design** (Floorplanning, Placement, CTS, Routing), **DRC/LVS**
Programming & Scripting: SystemVerilog, **Verilog**, **Python** (Basic), TCL, **C++**, MATLAB, SQL (Basic)
Operating Systems: Linux, Windows

ACADEMIC PROJECTS

ASIC Acceleration for GCN (Graph Convolution Neural Network) – RTL to GDSII (7nm PDK) — *Fall 2025*
- Developed testbenches and ran **functional verification** of convolution and pooling modules in **ModelSim**, validating correctness across multiple edge-case scenarios before synthesis.
- Ran **logic synthesis** in **Synopsys Design Compiler** under **timing, area, and power constraints** – evaluated **design trade-offs** across iterations to meet target performance.
- Drove complete **physical design flow** in **Cadence Innovus** (floorplanning, placement, CTS, routing) and achieved clean **DRC/LVS signoff**, closing the full **ASIC design flow** from RTL through sign-off.

2-Bit Adder – RTL to GDSII Implementation (ASAP 7nm FinFET, Verilog) — *Fall 2025*
- Designed a 2-bit adder in **Verilog** and built **functional verification testbenches** in ModelSim, ensuring correct behavior under all input combinations.
- Synthesized RTL using **Synopsys Design Compiler** and ran **post-synthesis verification** to confirm gate-level equivalence.
- Completed full **APR flow** in **Cadence Innovus** and validated post-route **DRC/LVS**, closing the loop from RTL through silicon-ready signoff.

4-Bit Adder Design and Implementation (SAED 32nm, Cadence Virtuoso) — *Spring 2025*
- Designed schematic and **transistor-level implementation** of a 4-bit mirror adder in **Cadence Virtuoso**, integrating 1-bit adder cells and inverters.
- Analyzed **worst-case delay, average power, and layout area** to evaluate **design trade-offs** and identify performance bottlenecks.
- Achieved **DRC/LVS clean** physical layout, demonstrating end-to-end closure from schematic through physical verification.

Delay Optimization of Logic Path (32nm PDK) — *Spring 2025*
- Designed a static CMOS logic path using standard cells and applied logical techniques to minimize **critical path delay**.
- Analyzed **timing reports** and **parasitic extraction** results to evaluate delay, power, and layout-dependent effects.
- Wrote **scripted automation** to compare performance across sizing iterations and validate optimization decisions against design targets.

PROFESSIONAL EXPERIENCE

Teleindia Networks Pvt Ltd *GET Trainee – Cloud (IT Team)* — *July 2024 – December 2024*
- Managed and configured **Windows and Linux virtual machines**, building proficiency in Linux-based environments used across EDA and verification workflows.
- Developed the DataGraph web application using PHP, HTML, CSS, and JavaScript, applying **structured debugging and version-controlled development practices**.
- Deployed and maintained applications on **cloud-based servers**, gaining exposure to **regression management and CI/CD pipelines**.